

```

11. except ImportError:
12.     print('This program requires the bext module, which you')
13.     print('can install by following the instructions at')
14.     print('https://pypi.org/project/Bext/')
15.     sys.exit()
16.
17. # Set up the constants:
18. WIDTH = 79
19. HEIGHT = 22
20.
21. TREE = 'A'
22. FIRE = 'W'
23. EMPTY = ' '
24.
25. # (!) Try changing these settings to anything between 0.0 and 1.0:
26. INITIAL_TREE_DENSITY = 0.20 # Amount of forest that starts with trees.
27. GROW_CHANCE = 0.01 # Chance a blank space turns into a tree.
28. FIRE_CHANCE = 0.01 # Chance a tree is hit by lightning & burns.
29.
30. # (!) Try setting the pause length to 1.0 or 0.0:
31. PAUSE_LENGTH = 0.5
32.
33.
34. def main():
35.     forest = createNewForest()
36.     bext.clear()
37.
38.     while True: # Main program loop.
39.         displayForest(forest)
40.
41.         # Run a single simulation step:
42.         nextForest = {'width': forest['width'],
43.                       'height': forest['height']}
44.
45.         for x in range(forest['width']):
46.             for y in range(forest['height']):
47.                 if (x, y) in nextForest:
48.                     # If we've already set nextForest[(x, y)] on a
49.                     # previous iteration, just do nothing here:
50.                     continue
51.
52.                 if ((forest[(x, y)] == EMPTY)
53.                     and (random.random() <= GROW_CHANCE)):
54.                     # Grow a tree in this empty space.
55.                     nextForest[(x, y)] = TREE
56.                 elif ((forest[(x, y)] == TREE)
57.                       and (random.random() <= FIRE_CHANCE)):
58.                     # Lightning sets this tree on fire.
59.                     nextForest[(x, y)] = FIRE
60.                 elif forest[(x, y)] == FIRE:
61.                     # This tree is currently burning.
62.                     # Loop through all the neighboring spaces:
63.                     for ix in range(-1, 2):
64.                         for iy in range(-1, 2):

```